Multiple imputation with mice

Inference lab using the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Run multiple imputation by chained equations with `mice`.
- Diagnose convergence with trace and density plots.
- Pool regression coefficients across imputations using Rubin's rules.

Prerequisites

The MCAR/MAR/MNAR lab.

Background

Multiple imputation (MI) replaces each missing value with several plausible values drawn from a predictive distribution, producing m completed datasets. Each completed dataset is analysed separately; the results are pooled using Rubin's rules so that the standard errors reflect both the within-imputation uncertainty (ordinary sampling variance) and the between-imputation uncertainty (the variance of the imputations themselves).

`mice` implements MI by chained equations: for each incomplete variable, it fits a conditional model given the others and draws imputations from the posterior predictive distribution of that model. The procedure iterates until the imputations stabilise. The standard diagnostics are trace plots (should mix and not drift) and density plots comparing observed and imputed values (should overlap but need not match exactly).

The inclusion of the outcome in the imputation model is not optional. Omitting it biases the imputations toward the null and attenuates the estimated effect in the pooled analysis.

Setup

```
library(tidyverse)
library(mice)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

In the `mice::nhanes` teaching dataset, the cholesterol outcome (`chl`) is related to age and BMI (`bmi`). We will estimate that relationship under MI.

2. Visualise

```
md.pattern(nhanes, plot = FALSE)
```

```
  mutate(missing_chl = is.na(chl)) |>
  ggplot(aes(bmi, chl, colour = missing_chl)) +
  geom_point(size = 2) +
  labs(x = "BMI", y = "Cholesterol", colour = "chl missing?")
```

3. Assumptions

MAR conditional on the variables in the imputation model; a compatible imputation model (predictive mean matching by default); enough imputations (here $m = 10$) to stabilise the pooled standard errors.

4. Conduct

```
            printFlag = FALSE, seed = 42)
imp
```

```
plot(imp)
```

```
densityplot(imp)
```

```
pooled <- pool(fit)
summary(pooled, conf.int = TRUE)
```

5. Concluding statement

> Multiple imputation of the `nhanes` dataset ($m = 10$, predictive mean matching) gave a pooled coefficient for BMI on cholesterol of `round(summary(pooled)$estimate[summary(pooled)$term == "bmi"], 2)` (see pooled table above), with standard errors reflecting both within- and between-imputation uncertainty through Rubin's rules.

Common pitfalls

- Imputing the outcome separately from the covariates with incompatible models.
- Using $m = 5$ when the fraction of missing information is high.
- Pooling means or SDs by hand instead of `pool()`.
- Dropping the outcome from the imputation model because it "feels like cheating".

Further reading

- van Buuren S, Groothuis-Oudshoorn K (2011), *mice: Multivariate Imputation by Chained Equations in R*.
- van Buuren S (2018), *Flexible Imputation of Missing Data*, ch. 4–6.
- White IR, Royston P, Wood AM (2011), *Multiple imputation using chained equations: issues and guidance for practice*.

Session info

```

```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)